Eric Balch

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Education

University of California, Los Angeles — Sept. 2024 – June 2028
Bachelor of Science in Mechanical Engineering — *GPA: 3.98/4.0*
- Extracurriculars: Bruin Formula SAE Racing, Bionics Lab, Tau Beta Pi

Experience

Mechanical Engineering Intern — June 2026 – Present
ANELLO Photonics Inc. — *Santa Clara, CA*
- Electromechanical design, development, assembly, & testing of Inertial Navigation System calibration fixtures
- Selection and integration of high-precision motors and encoders for repeatable actuation
- Design breakout cable tester to validate harness functionality and streamline manufacturing quality control
- Increased unit calibration throughput by 250% while maintaining 2 mG performance accuracy

Drivetrain Lead & Responsible Engineer — Oct. 2024 – Present
Bruin Formula SAE Racing Drivetrain Team — *Los Angeles, CA*
- Directing drivetrain architecture, leading team of 10 through full design cycle
- Design (SolidWorks) front/rear sprockets and motor shaft w/ Design-for-Manufacture (DFM) emphasis
- Iteratively FEA-test (ANSYS Workbench) part geometry to optimize slotting & reduce weight by >10% while withstanding extreme mechanical stress
- Develop GD&T drawings & communicate specs to local sponsors for outsourcing component manufacturing
- Create and execute (F360 CAM) CNC toolpaths for manufacturing of motor shaft and motor mounts, reducing manufacturing costs by >25%
- Reduced manufacturing time for differential and motor mounts by >20% by transitioning from soft jaws to tabbing
- Designed (SolidWorks) 80/20 assembly to stress test motor-to-sprocket chain (Up to 4500 N)

Undergraduate Researcher — Nov. 2025 – Present
UCLA Bionics Laboratory — *Los Angeles, CA*
- Design (SolidWorks) endoscope with independently controlled camera and surgical accessories to improve polyp resection and ease of use for surgeons
- Focus on the distal component of the device, creating the housing and operational mechanism for the accessory attachment point
- Rapid prototyping of designs using 3D printed components & off-the-shelf tubing & control wires

Biomechanical Engineering Intern — June 2023 – Sept. 2023
Stanford University — *Stanford, CA*
- Conducted modeling (segmentation, model building, meshing) for computational fluid dynamics (CFD) simulations to understand coronary artery disease and physiology
- Modeled pipeline for cardiovascular anatomic models of patient-specific coronary bypass procedures (e.g., grafts) to analyze changing hemodynamics of different attachment locations
- Created all segmentations and ran CFD simulations to evaluate standard graft with resistor and complex boundary conditions (e.g., coronary BCs) to consider the effects of multiple stenoses

Lifeguard and Swim Instructor — June 2021 – Sept. 2025
City of Mountain View — *Mountain View, CA*
- Head Guard leadership role supervising a staff of up to 5 lifeguards, CPR & water rescue certified

Relevant Projects

Team Lead & Lead Designer, Engineering Lab Course — Sept. 2022 – June 2024
- Designed and built winning wireless car parking lot sensors & display for CTE Capstone project
- Designed and built winning robots for two Vex Robotics timed competitive challenges
- Designed and built hydraulic arm to perform timed competitive precision challenges
- Coordinated team member contributions and presented product design results to class

Technical Skills

Software: SolidWorks, Fusion 360, ANSYS, MS Office, Gemini, MATLAB, Python, Java
Tools: Lathe & Mill (CNC & Manual), Drill Press, Band Saw, 3D Printer, Laser Cutter, Soldering